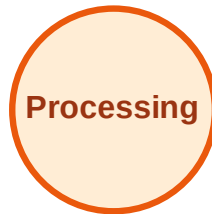


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

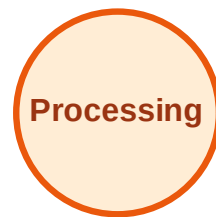
Visited: (none)



BFS Traversal

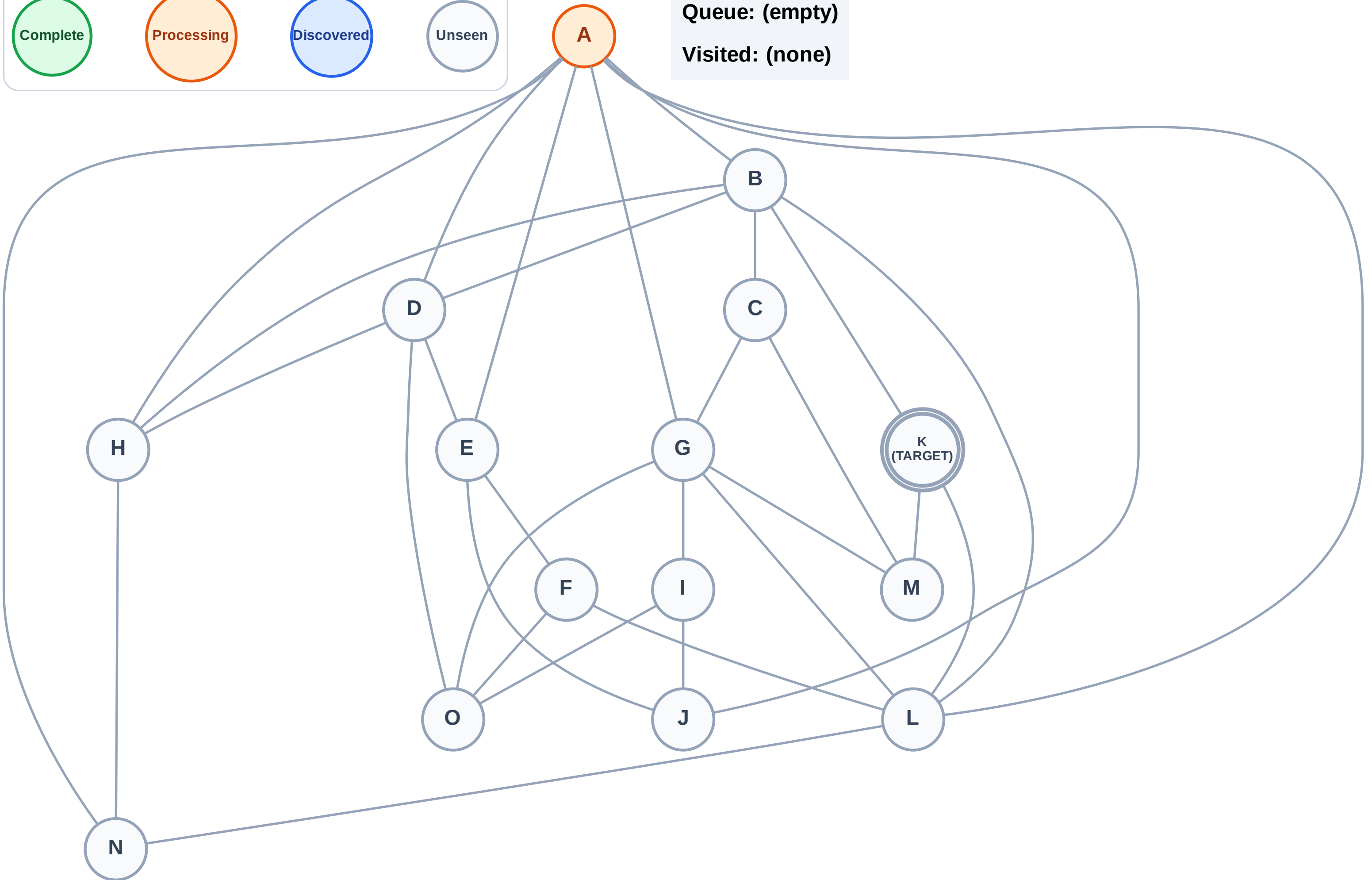
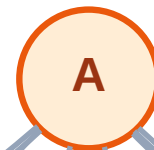
Step 2: Process node A

Legend



Queue: (empty)

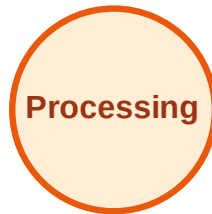
Visited: (none)



BFS Traversal

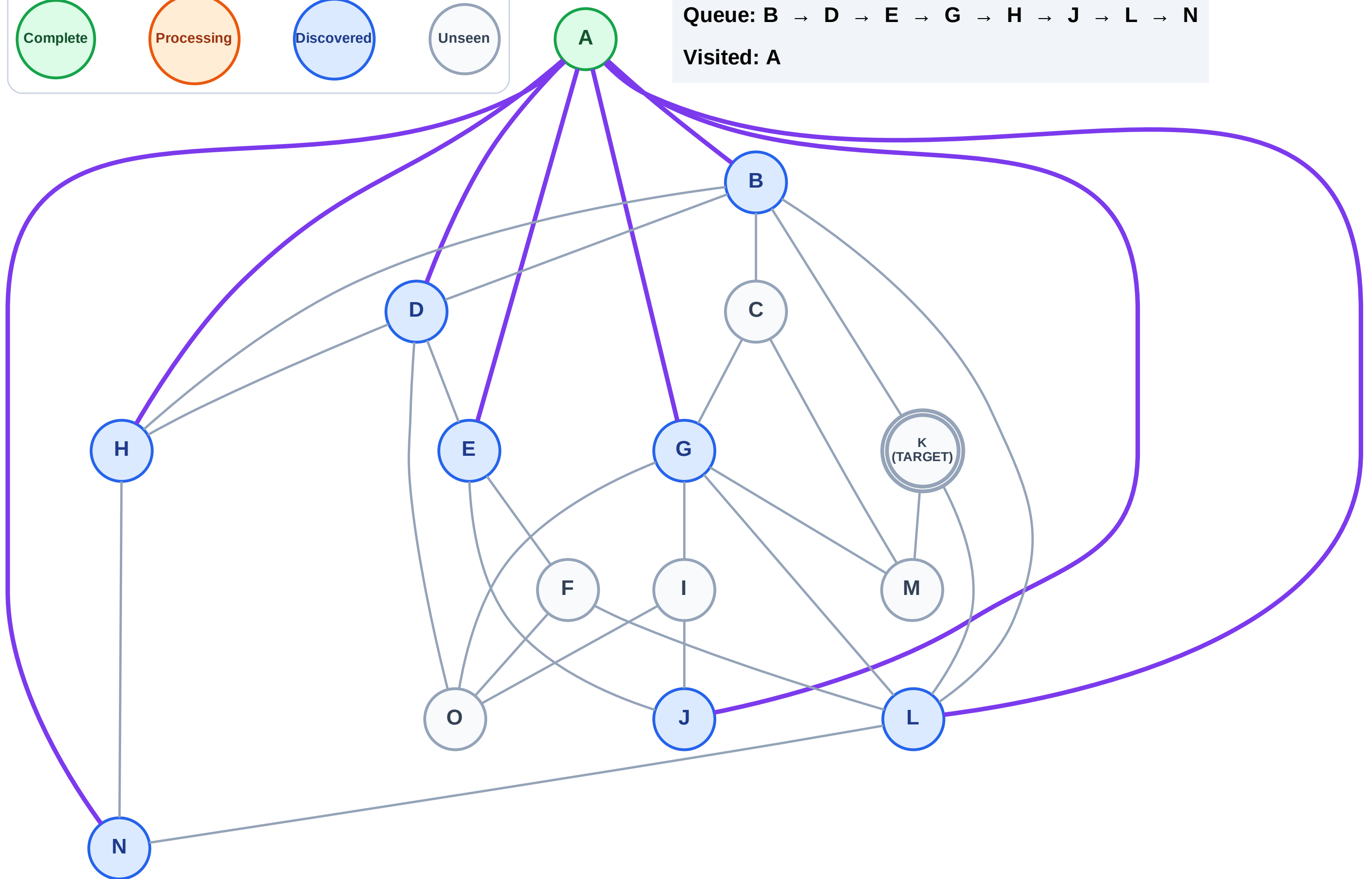
Step 3: Discover B, D, E, G, H, J, L, N from A

Legend



Queue: B → D → E → G → H → J → L → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

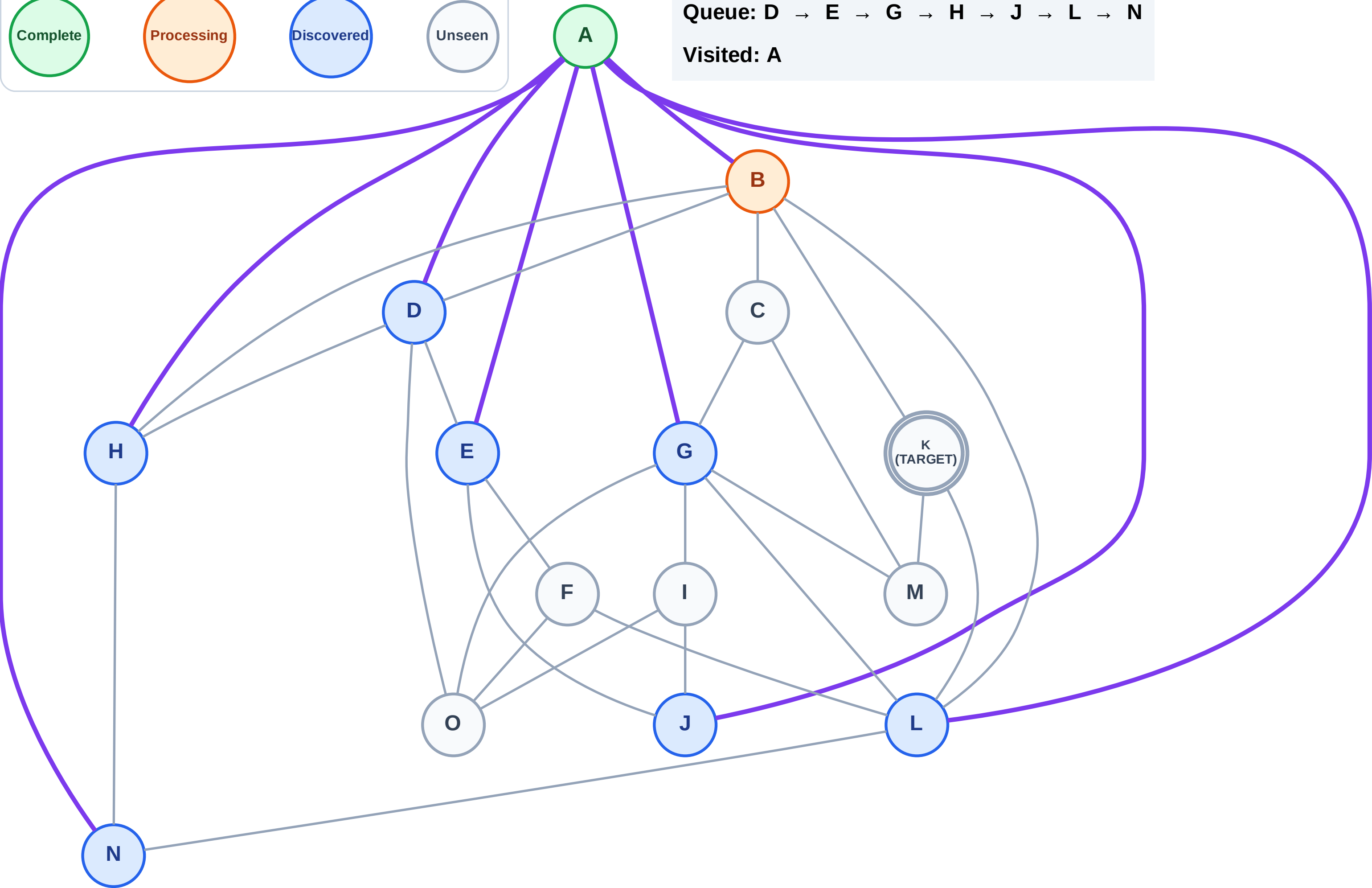
Processing

Discovered

Unseen

Queue: D → E → G → H → J → L → N

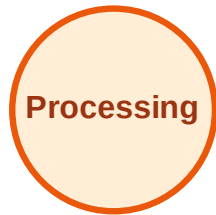
Visited: A



BFS Traversal

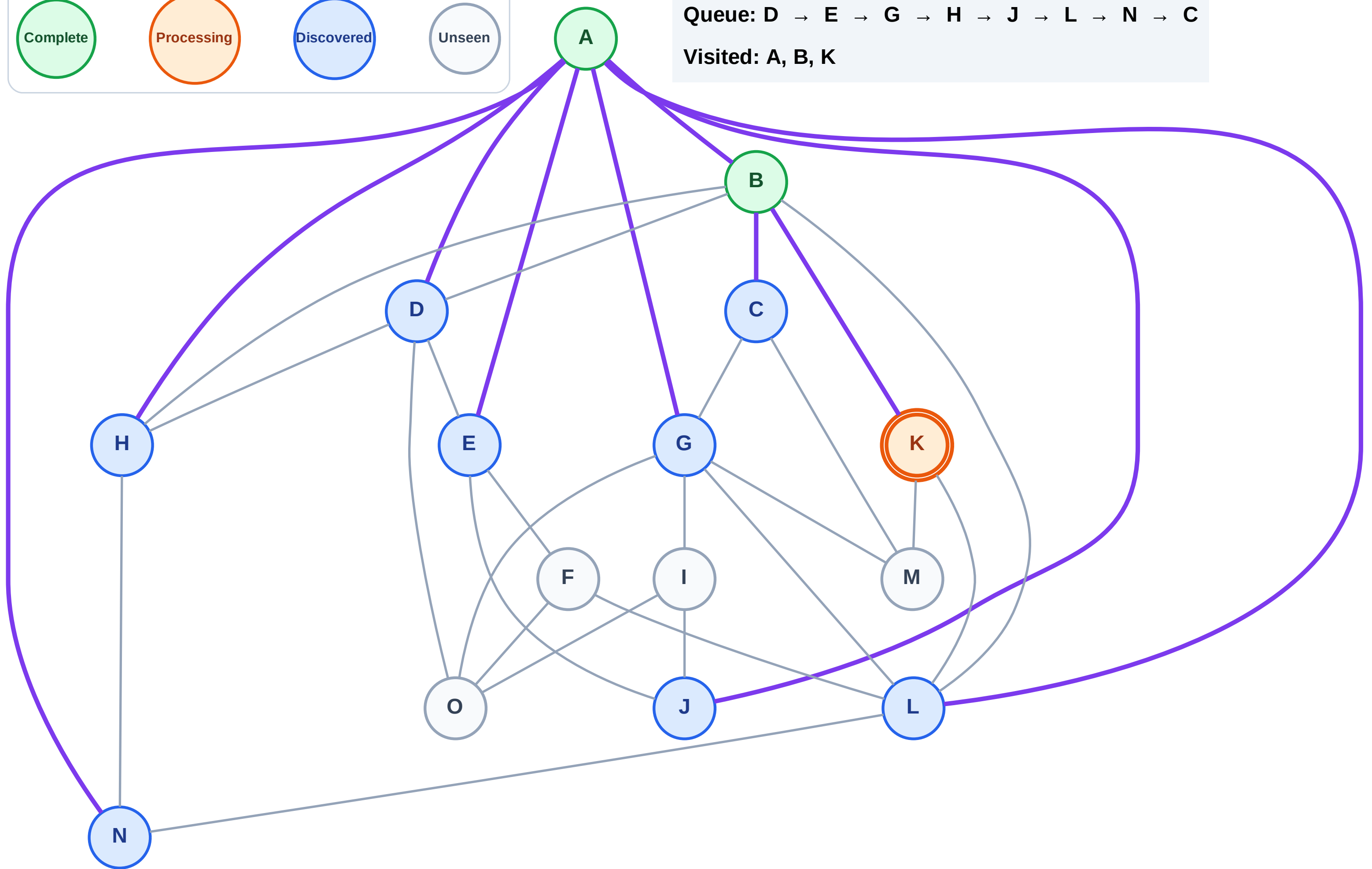
Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Queue: D → E → G → H → J → L → N → C

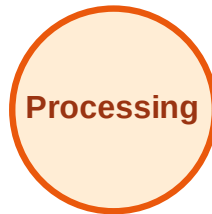
Visited: A, B, K



DFS Traversal

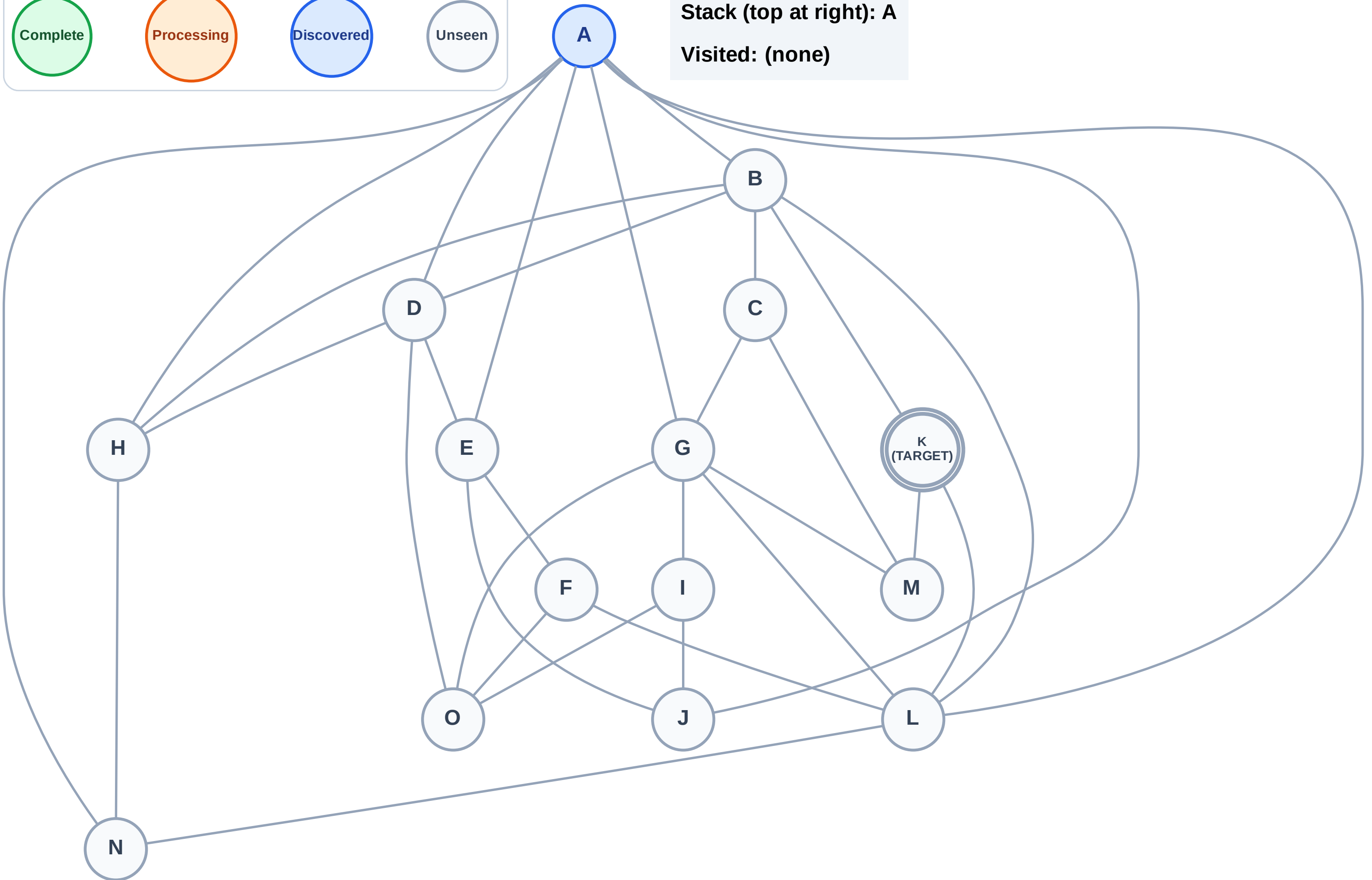
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

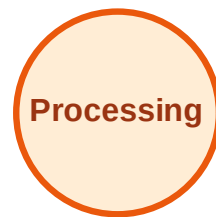
Visited: (none)



DFS Traversal

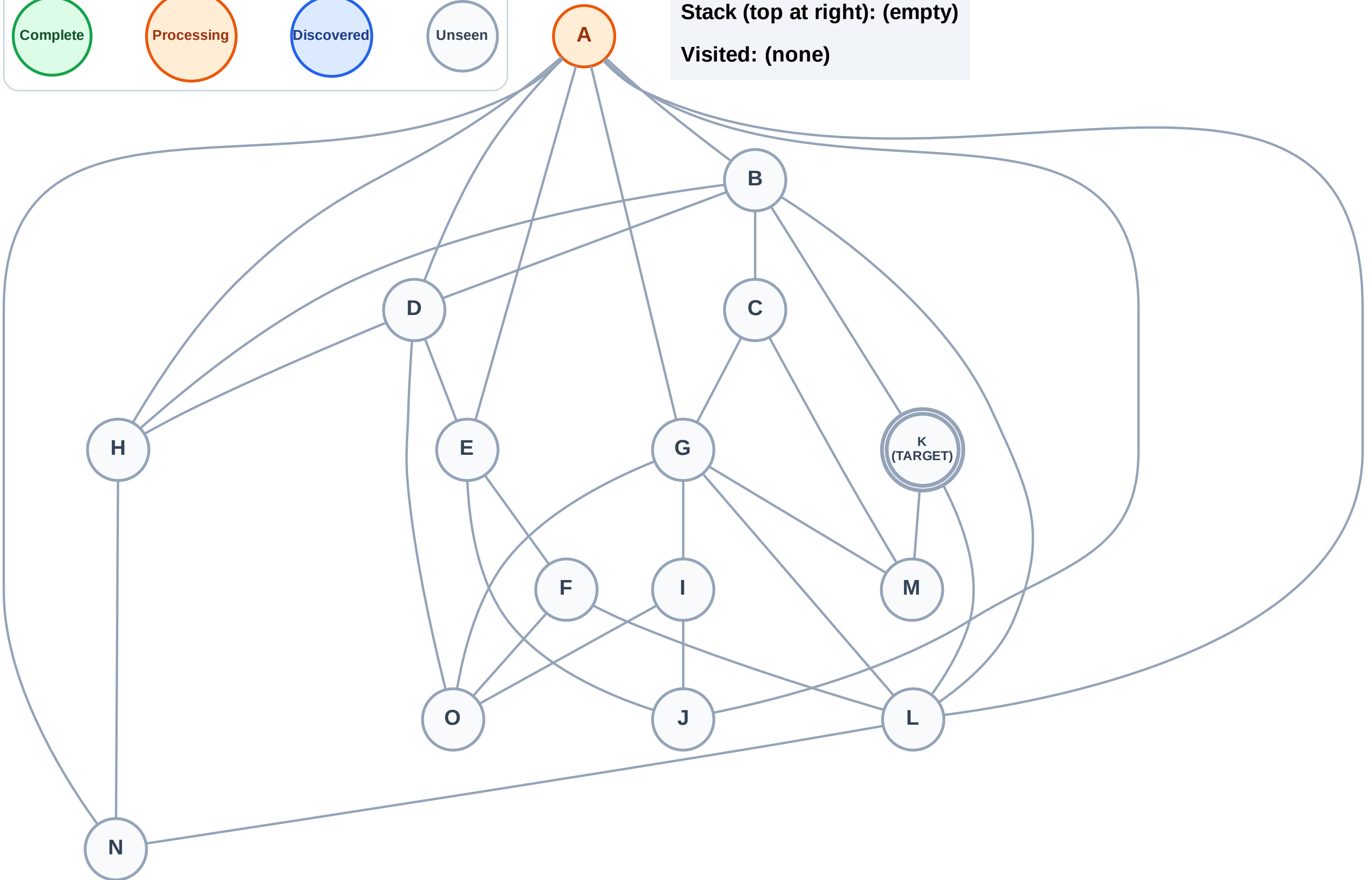
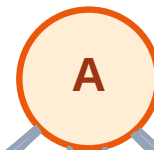
Step 2: Process node A

Legend



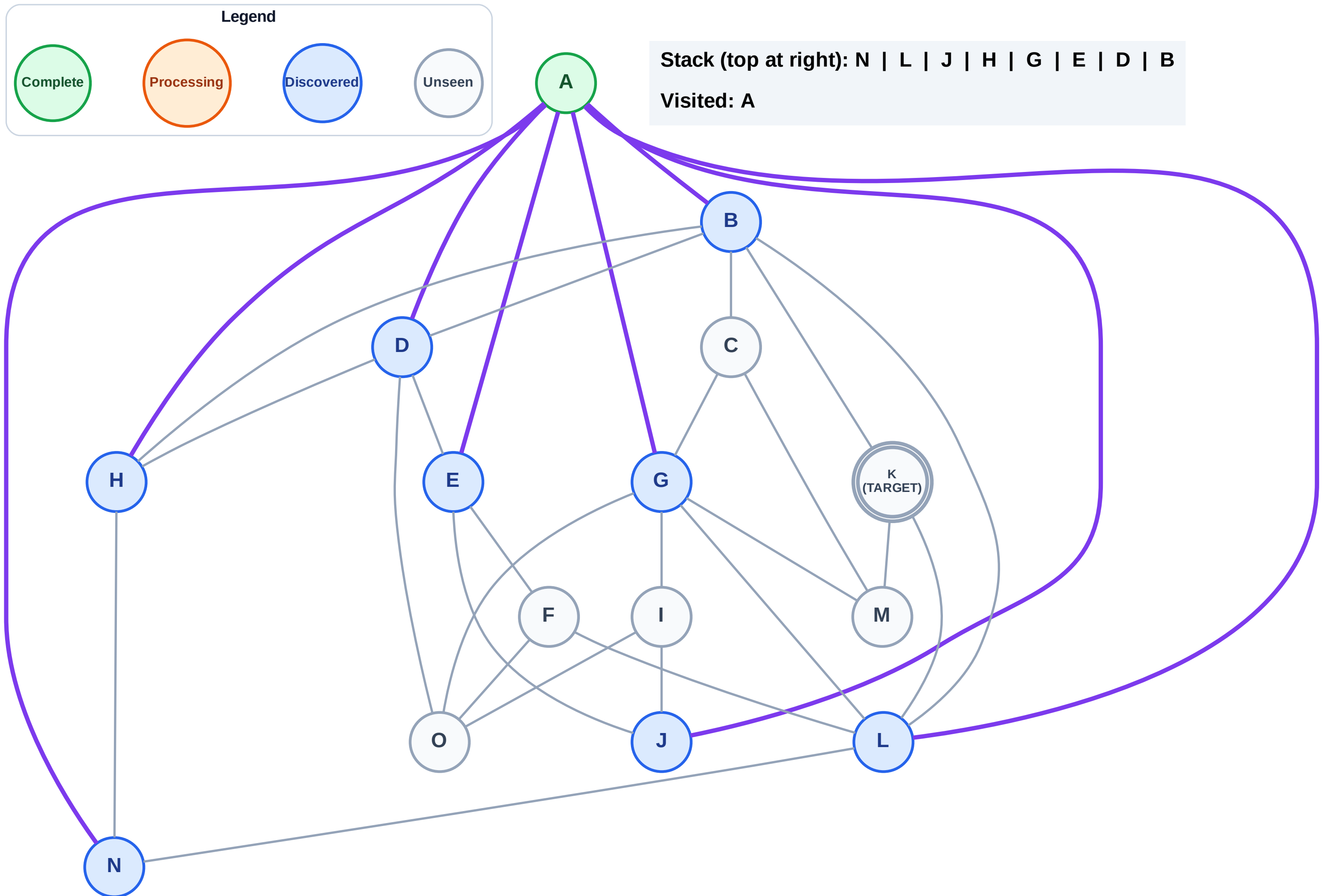
Stack (top at right): (empty)

Visited: (none)



Step 3: Discover N, L, J, H, G, E, D, B from A

Step 3: Discover N, L, J, H, G, E, D, B from A



DFS Traversal

Step 4: Process node B

Legend

Complete

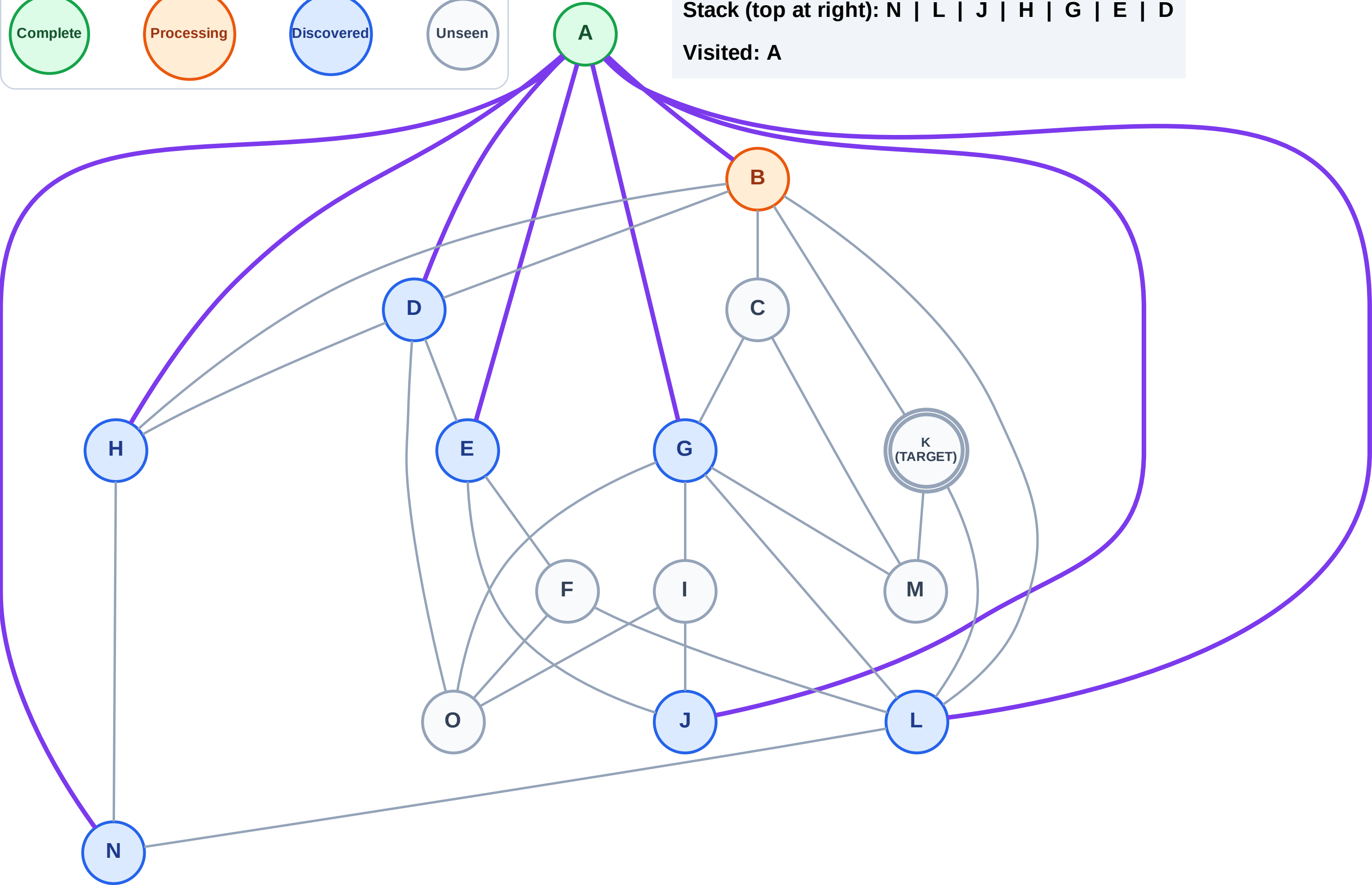
Processing

Discovered

Unseen

Stack (top at right): N | L | J | H | G | E | D

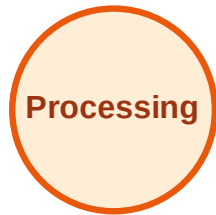
Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Stack (top at right): N | L | J | H | G | E | D

Visited: A, B, K

